Statistics Physics

Chengyu Jin

March 24, 2026

Contents

1 Introduction **2**

1.0.1 Design of the Canonical Ensemble 3

1.1 Connection with Thermodynamics 8

1.2 Partition Function. Classical and Quantum Formalism 15

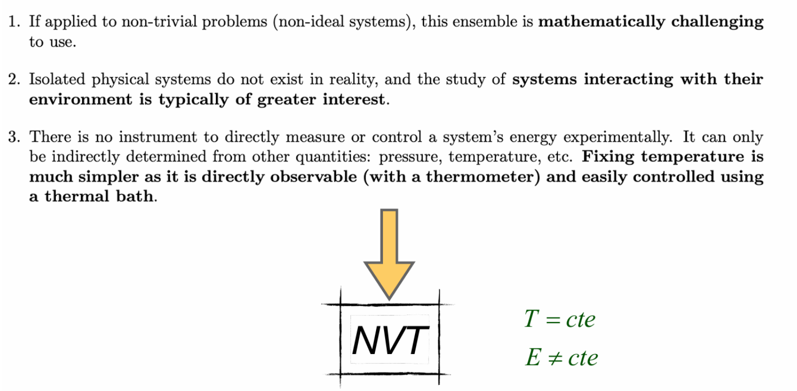
1.3 Application to Ideal Classical Systems: Boltzmann Statistics 18

1.3.1 Case 1. Ideal systems of identical and distinguishable particles 20

1.3.2 Case 2. Ideal systems of identical and indistinguishable particles 22

Chapter 1

Introduction

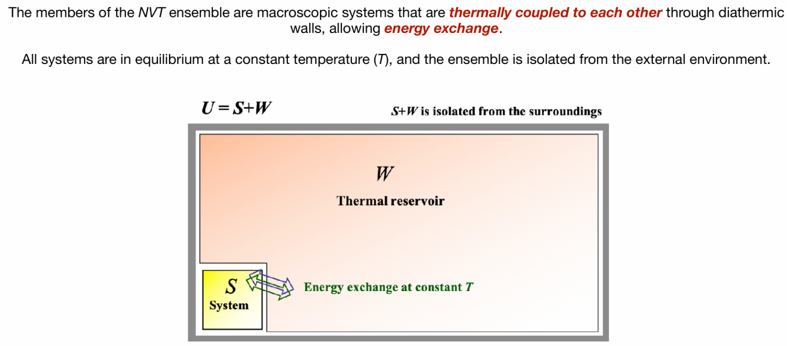


In the previous chapter, we have learnt that the microcanonical (NVE) ensemble is employed to study systems with a fixed total energy and no energy exchange with the surroundings. As a result, it characterises the statistical behaviour of an isolated system, considering the multiplicity of energy states accessible to the system for a given fixed energy. We have already mentioned that the NVE ensemble presents significant drawbacks:

1. If applied to non-trivial problems (non-ideal systems), this ensemble is **mathematically challenging** to use.

2. Isolated physical systems do not exist in reality, and the study of **systems interacting with their environment** is typically of greater interest.

3. There is no instrument to directly measure or control a system's energy experimentally. It can only be indirectly determined from other quantities: pressure, temperature, etc. **Fixing temperature is much simpler as it is directly observable (with a thermometer) and easily controlled using a thermal bath.**

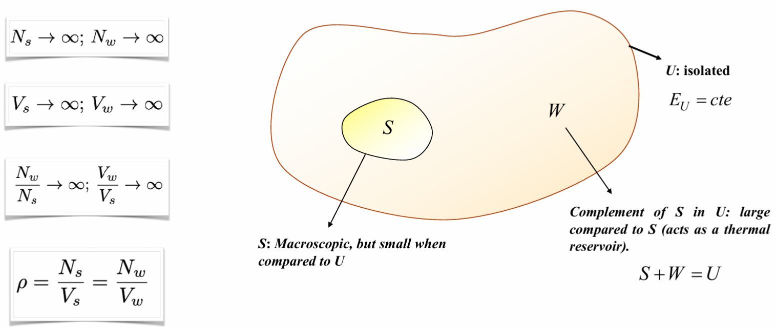


In this chapter, we introduce and describe the **canonical ensemble**, also indicated as NVT ensemble, which deals with systems that can exchange energy with a larger reservoir (bath) at a constant temperature. In the NVT ensemble, systems are allowed to interact with an external reservoir, enabling energy exchanges.

This interaction permits the system to explore various energy states within the constraints of a fixed temperature. The resulting fluctuations among energy states contribute to the statistical distribution of energy across accessible states, allowing for a probabilistic description of the system.

The members of the NVT ensemble are macroscopic systems thermally coupled through diathermic walls, allowing energy exchange between them. All systems are in equilibrium at a constant temperature (T), and the ensemble is isolated from the external surroundings as schematically illustrated in Fig. 5.1.

1.0.1 Design of the Canonical Ensemble



With reference to Fig. 5.1, U is an isolated *supersystem* with E_u constant. The system S is a region of U that is not strongly fluctuating and is macroscopic, but small compared to U. The complement of S in U, indicated by W, is large compared to S and acts as a thermal bath. We also know that $1 \ll N_s \ll N_w$, where N_s and N_w are, respectively, the number of particles in S and in W. Similarly, $1 \ll V_s \ll V_w$, with V_s and V_w the volume occupied, respectively, by S and W. In particular, the following relations hold:

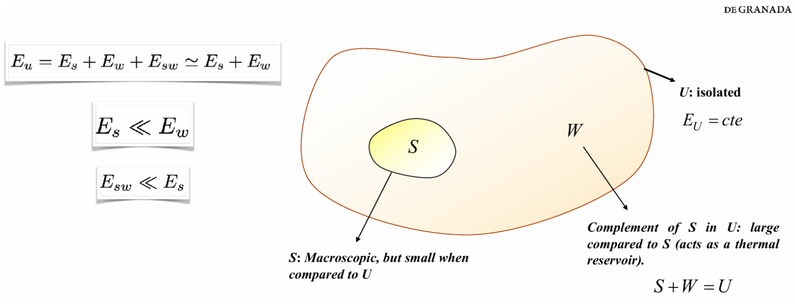
$$N_s \to \infty; N_w \to \infty; V_s \to \infty; V_w \to \infty \tag{5.1}$$

and, in the thermodynamic limit, the size of the bath is infinitely larger than that of the system of interest. Mathematically, this condition is expressed as

$$\frac{N_w}{N_s} \to \infty; \frac{V_w}{V_s} \to \infty \tag{5.2}$$

We also assume that the density of S equals the density of W:

$$\rho = \frac{N_s}{V_s} = \frac{N_w}{V_w} \tag{5.3}$$



Finally, the internal energy of the supersystem (universe) is related to that of the system and the bath as follows:

$$E_u = E_s + E_w + E_{sw} \simeq E_s + E_w \tag{5.4}$$

where $E_s \ll E_w$ and $E_{sw} \ll E_s$ are negligible. As a consequence, given the negligible interaction between S and W, one can write the following relation:

$$E_m = E_m(N_s, V_s) \tag{5.5}$$

where E_m, which depends on the number of particles and the volume of S, denotes the energy of an energy state m in which S can be found. If the energy of S is E_m, then the energy of the thermal bath W must lie between

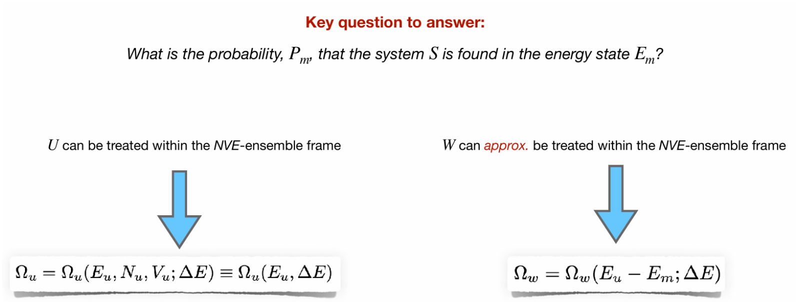
$$E_w = E_u - E_m \tag{5.6}$$

and

$$E_w + \Delta E. \tag{5.7}$$

This energy interval of *uncertainty* arises from the fact that energy cannot be measured exactly, as already noted in the discussion of the microcanonical ensemble in Chapter 4. From now on, to simplify the notation, we will use N and V to refer, respectively, to the number of particles and volume of S.

Partition Function

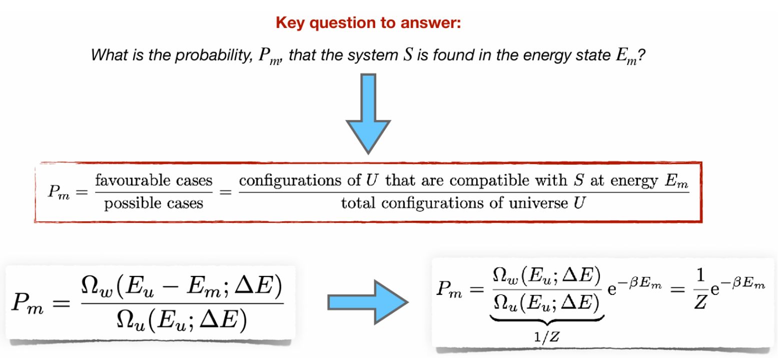


Now that the thermodynamic system of interest has been defined, the question we aim to address is: *What is the probability, P_m, that the system S is found in the energy state E_m?* To answer this question, we note that the supersystem U is isolated, with E_u, N_u, and V_u constant. This implies that U can be treated within the microcanonical ensemble framework. In particular,

$$\Omega_u = \Omega_u(E_u, N_u, V_u; \Delta E) \equiv \Omega_u(E_u, \Delta E). \tag{5.8}$$

Strictly speaking, the thermal bath W, with energy $E_u - E_m$, is not isolated, as it can exchange energy with S. Nevertheless, since $E_m \ll E_u$, we can neglect the effect of S on W and assume, to a good approximation, that W is also effectively isolated:

$$\Omega_w = \Omega_w(E_u - E_m; \Delta E). \tag{5.9}$$



We emphasize that Ω_w represents the number of configurations (states) in the universe U that are compatible with an energy E_m for S, where E_w is such that the total energy falls within the range $[E_u, E_u + \Delta E]$. Additionally, since W can be considered an isolated system (as its interactions with S can be neglected), it can be treated within the microcanonical ensemble framework, implying that all its possible configurations (states) are equally probable. In light of these preliminary considerations, let's calculate this probability:

$$P_m = \frac{\text{favourable cases}}{\text{possible cases}} = \frac{\text{configurations of } U \text{ that are compatible with } S \text{ at energy } E_m}{\text{total configurations of universe } U} \tag{5.10}$$

equivalently

$$P_m = \frac{\Omega_w(E_u - E_m; \Delta E)}{\Omega_u(E_u; \Delta E)} \tag{5.11}$$

Given that $E_m \ll E_u$, we can develop a power series of Ω_w. However, since Ω is generally a function that increases exponentially with energy, the convergence of the series will be very slow. An alternative is expanding $\ln \Omega_w$:

$$\ln \Omega_w(E_w; \Delta E) = \ln \Omega_w(E_u; \Delta E) + \underbrace{\left(\frac{\partial \ln \Omega_w}{\partial E}\right)_{E=E_u}}_{\beta} (E_w - E_u) + \cdots \simeq \ln \Omega_w(E_u; \Delta E) - \beta E_m \tag{5.12}$$

Rearranging, we obtain

$$\Omega_w(E_u - E_m; \Delta E) \simeq \Omega_w(E_u; \Delta E) e^{-\beta E_m} \tag{5.13}$$

and substituting in Eq. 6.20, we get

$$P_m = \underbrace{\frac{\Omega_w(E_u; \Delta E)}{\Omega_u(E_u; \Delta E)}}_{1/Z} e^{-\beta E_m} = \frac{1}{Z} e^{-\beta E_m} \tag{5.14}$$

where Z is the **canonical partition function**. The best way to calculate Z is not to use Eq. 5.14, but to recall the normalisation condition of probability:

$$\sum_m P_m = \sum_m \frac{1}{Z} e^{-\beta E_m} = \frac{1}{Z} \sum_m e^{-\beta E_m} = 1 \tag{5.15}$$

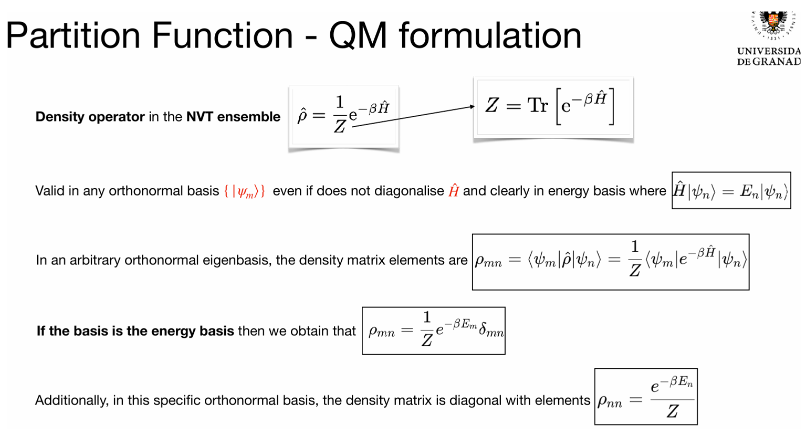
It follows that

$$Z = \sum_m e^{-\beta E_m} \tag{5.16}$$

and equivalently

$$Z = \sum_n \Omega(E_n) e^{-\beta E_n} \tag{5.17}$$

where n indicates the number of energy levels. Since each energy level E_n can have multiple states associated with it, Z can be expressed either as a sum over individual states or as a sum over energy levels weighted by their degeneracy $\Omega(E_n)$, as in Eq. 5.17. The partition function Z encodes how the probabilities are **partitioned** among the different states, based on their individual energies[1]. It represents the sum over all the energy states of S compatible with the same NVT-macrostate or, equivalently, the sum over all the energy levels observed in S. The partition function is much easier to calculate than Ω and is valid for systems that are more general than the isolated one.



The quantum formulation of the canonical ensemble is represented by the density matrix ρ_{mn}, which is given by:

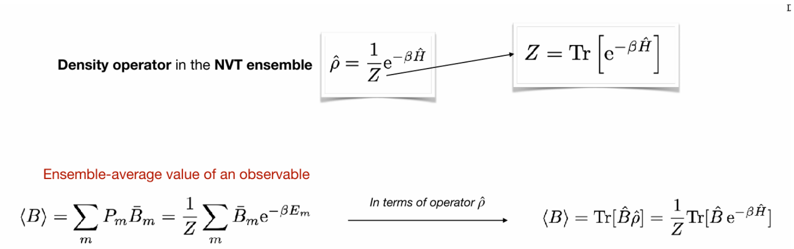
$$\rho_{mn} = \frac{1}{Z} e^{-\beta E_m} \delta_{mn}. \tag{5.18}$$

In general, expectation values are calculated as

$$\langle B \rangle = \sum_m P_m B_m = \frac{1}{Z} \sum_m B_m e^{-\beta E_m}, \tag{5.19}$$

where $B_m = \langle \phi_m | \hat{B} | \phi_m \rangle$ is the expectation value of $\hat{B}$ in the eigenstate $|\phi_m\rangle$. One of the key advantages of the canonical ensemble is that it can be applied using any orthonormal basis $|\psi_m\rangle$, even if it does not diagonalize $\hat{H}$. This implies that solving Schrödinger's equation explicitly is not necessary. In terms of the density operator, we define:

$$\hat{\rho} = \frac{1}{Z} e^{-\beta \hat{H}}. \tag{5.20}$$



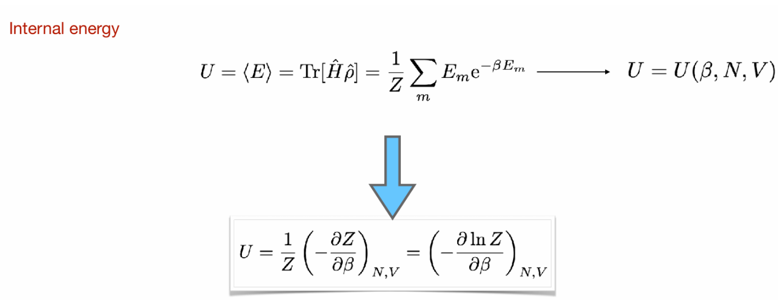
It follows that

$$\langle B \rangle = \mathrm{Tr}[\hat{B}\hat{\rho}] = \frac{1}{Z} \mathrm{Tr}\left[\hat{B} e^{-\beta \hat{H}}\right]. \tag{5.21}$$

Thus, we can use any orthonormal basis of the Hilbert space to perform numerical calculations:

$$Z = \mathrm{Tr}\left[e^{-\beta \hat{H}}\right]. \tag{5.22}$$

1.1 Connection with Thermodynamics



Thermodynamics begins with a few empirical observations, which are organised and axiomatised in the form of four laws. From these laws, thermodynamics establishes relationships between several properties that characterise the thermal behaviour of a macroscopic system. However, thermodynamics presents an important drawback:

it does not allow the calculation of absolute values for any property. For example, if we know C_V, we can determine C_P and viceversa; thermodynamics provides the relationship between them but not their specific values. Statistical Physics can resolve this inconvenience; specifically, the canonical ensemble provides a model for thermal equilibrium. More specifically, Statistical Physics is capable of obtaining thermodynamic (macroscopic) properties in terms of microscopic magnitudes of molecules. In the case of the canonical ensemble, if we know the energy spectrum $\{E_n\}$, we can deduce the thermodynamic variables. Similarly, Statistical Physics allows for the calculation of microscopic properties from macroscopic observations (for example, intermolecular interactions).

Thermodynamic variables can be classified into three groups:

- **External Parameters**, such as N (number of particles), V (volume), etc. Fixed by the experimenter or the system's environment.
- **Mechanical Properties**, such as p (pressure), U (internal energy), etc. They are averages of microscopic dynamic functions.
- **Thermal Properties**, such as T (temperature), S (entropy). They do not have a microscopic equivalent (at the particle level). They can only be defined at the macroscopic level. They are collective properties whose value is determined by the entire ensemble.

The first connection is provided by the internal energy:

$$U = \langle E \rangle = \mathrm{Tr}[\hat{H}\hat{\rho}] = \frac{1}{Z}\sum_m E_m e^{-\beta E_m} \tag{5.23}$$

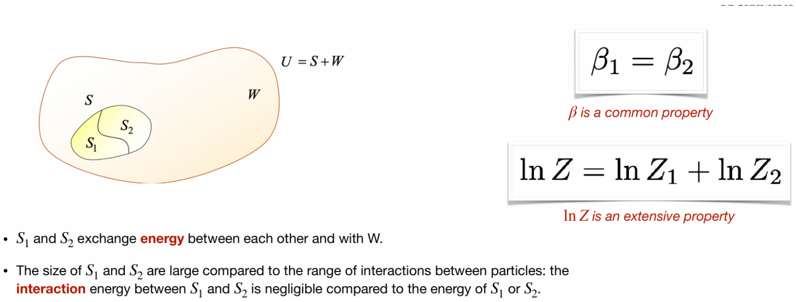
We note that $U = U(\beta, N, V)$, although we still have to see what β is. Now, since the system can exchange energy with the environment (in equilibrium), E is no longer fixed but can fluctuate. The internal energy is its mean value $U = \langle E \rangle$. It is possible to re-write Eq. 6.28 in a more practical fashion:

$$U = \sum_m E_m P_m = \frac{1}{Z}\sum_m E_m e^{-\beta E_m} = \frac{1}{Z}\sum_m \left(-\frac{\partial}{\partial\beta}e^{-\beta E_m}\right) = \frac{1}{Z}\left(-\frac{\partial}{\partial\beta}\right)\sum_m e^{-\beta E_m} \tag{5.24}$$

Equivalently:

$$U = \frac{1}{Z}\left(-\frac{\partial Z}{\partial\beta}\right)_{N,V} = -\left(\frac{\partial \ln Z}{\partial\beta}\right)_{N,V} \tag{5.25}$$

To find out further connections with Thermodynamics, specifically that β is closely related to the temperature, we now consider the configuration given in Fig. 5.2.



Systems S_1 and S_2 exchange energy between themselves and with W and the three systems are at thermal equilibrium with each other. We will assume that the sizes of S_1 and S_2 are large compared to the range of interactions between particles: the interaction energy between S_1 and S_2, although physically significant in ensuring the exchange of energy, is negligible compared to the energy of S_1 or S_2. What is the probability, $P_{n,m}$, of finding system S_1 in the energy state E_{1n} and S_2 in E_{2m}? To answer this question, we have two options:

- If we consider S as a sum of S_1 and S_2, then

$$P_{n,m} = P_n^{(1)} P_m^{(2)} = \frac{1}{Z_1} e^{-\beta_1 E_{1,n}} \frac{1}{Z_2} e^{-\beta_2 E_{2,m}} \tag{5.26}$$

  This is basically the product of two independent probabilities, with, in principle, $\beta_1 \neq \beta_2$.

- If we can look at S as a unique system, then

$$P_{n,m} = \frac{1}{Z} e^{-\beta(E_{1,n}+E_{2,m})} \tag{5.27}$$

The probability $P_{n,m}$ must be the same regardless the option applied. This implies that

$$\frac{1}{Z_1 Z_2} e^{-\beta_1 E_{1,n}} e^{-\beta_2 E_{2,m}} = \frac{1}{Z} e^{-\beta(E_{1,n}+E_{2,m})} \tag{5.28}$$

Since this equation is true $\forall E_{1,n}$ and $\forall E_{2,m}$, then it follows that

$$\beta_1 = \beta_2 \tag{5.29}$$

Because thermodynamics asserts that two systems in equilibrium must have the same temperature, it must be that

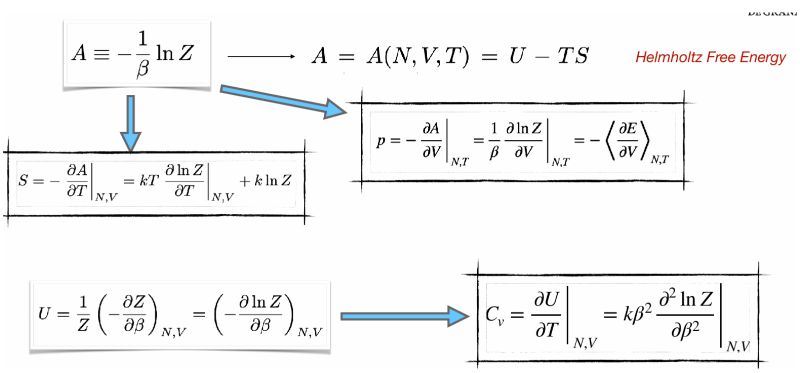
is an **intensive** property (in particular, $\beta = 1/kT$). We also see that

$$Z = Z_1 Z_2 \tag{5.31}$$

which implies that

$$\ln Z = \ln Z_1 + \ln Z_2 \tag{5.32}$$

is an additive or **extensive** property. Moreover, we see that β **is a common property**, being the same in all systems when thermal equilibrium is achieved.



By applying these two conclusions, we can define the following function

$$A \equiv -\frac{1}{\beta} \ln Z \tag{5.33}$$

which is also extensive as $A_{1+2} = A_1 + A_2$. What does A represent? To find that out, let's calculate the total derivative of βA, noting that $A = A(\beta, \{E_n\})$

$$d(\beta A) = \left.\frac{\partial(\beta A)}{\partial \beta}\right|_{\{E_n\}} d\beta + \sum_n \left.\frac{\partial(\beta A)}{\partial E_n}\right|_{\beta} dE_n \tag{5.34}$$

The first term in the right hand side of the above equation is equal to the internal energy (see also Eq. 5.25):

$$\left.\frac{\partial(\beta A)}{\partial \beta}\right|_{\{E_n\}} = -\left.\frac{\partial(\ln Z)}{\partial \beta}\right|_{\{E_n\}} = -\left.\frac{\partial(\ln Z)}{\partial \beta}\right|_{N,V} = U \tag{5.35}$$

The second term is a probability. Let's see why:

$$\left.\frac{\partial(\beta A)}{\partial E_n}\right|_\beta = -\left.\frac{\partial(\ln Z)}{\partial E_n}\right|_\beta = -\frac{1}{Z}\left.\frac{\partial Z}{\partial E_n}\right|_\beta = -\frac{1}{Z}\frac{\partial}{\partial E_n}\left(\sum_m e^{-\beta E_m}\right)_\beta = \beta\frac{e^{-\beta E_n}}{Z} = \beta P_n \tag{5.36}$$

Substituting these results in Eq. 5.34, we obtain

$$d(\beta A) = U d\beta + \beta \sum_n P_n dE_n \tag{5.37}$$

Since $U d\beta = d(\beta U) - \beta dU$, we can rewrite this equation as

$$d[\beta(U - A)] = \beta dU - \beta \sum_n P_n dE_n \tag{5.38}$$

We have now to understand what the summation in the above equation, that is $\sum_n P_n dE_n$, represents *thermodynamically*. To this end, we first note that dE_n is an energy change of the system following an **adiabatic volume change**[2]. Mathematically, this volume-driven energy change can be expressed as

$$E_n' = E_n + \left.\frac{\partial E_n}{\partial V}\right|_N dV \tag{5.39}$$

Therefore

$$\sum_n P_n dE_n = \underbrace{\sum_n P_n E_n'}_{U'} - \underbrace{\sum_n P_n E_n}_{U} = dU_{\text{adiabatic}} \tag{5.40}$$

The First Law of Thermodynamics (conservation of energy) states that $dU = Q + W$, where Q and W represent, respectively, the amount of heat and work exchanged between the system and its surroundings[3]. Because the process we are considering is adiabatic, the First Law reduces to $dU_{\text{adiabatic}} = W$. In particular,

$$dU_{\text{adiabatic}} = W = \sum_n P_n dE_n = \sum_n P_n \left.\frac{\partial E_n}{\partial V}\right|_N dV \tag{5.43}$$

Since the work exchanged as a result of a volume change is $W = -pdV$, we can rewrite the last equation as

$$\sum_n P_n \left.\frac{\partial E_n}{\partial V}\right|_N dV = -pdV \tag{5.44}$$

Rearranging, we obtain an expression for pressure:

$$p = -\sum_n P_n \left.\frac{\partial E_n}{\partial V}\right|_N \tag{5.45}$$

If we now substitute this result into Eq. 5.38, we get

$$d[\beta(U-A)] = \beta dU - \beta \sum_n P_n dE_n = \beta[dU - W] = \beta Q \tag{5.46}$$

It is interesting and useful to observe that, by multiplying Q by β, one can transform this inexact differential into the exact differential $d[\beta(U-A)]$. This implies that β is the *integrating factor* of Q[4]. If we now apply the Second Law of Thermodynamics $Q = TdS$, we obtain

$$d[\beta(U-A)] = \beta Q = \beta T dS \tag{5.47}$$

This implies that the property $A = A(N, V, T) = U - TS$, defined in Eq. 5.33, is the **Helmholtz free energy**. The total differential of $A(N, V, T)$ reads

$$dA = \left.\frac{\partial A}{\partial N}\right|_{V,T} dN + \left.\frac{\partial A}{\partial V}\right|_{N,T} dV + \left.\frac{\partial A}{\partial T}\right|_{N,V} dT = -d\left(\frac{1}{\beta}\ln Z\right). \tag{5.48}$$

It follows that

$$\mu = \left.\frac{\partial A}{\partial N}\right|_{V,T} = \frac{1}{Z}\sum_m \left(\left.\frac{\partial E_m}{\partial N}\right|_{V,T} e^{-\beta E_m}\right) = \left\langle \frac{\partial E}{\partial N}\right\rangle_{V,T} \tag{5.49}$$

$$p = -\left.\frac{\partial A}{\partial V}\right|_{N,T} = \frac{1}{\beta}\left.\frac{\partial \ln Z}{\partial V}\right|_{N,T} = -\frac{1}{Z}\sum_m \left(\left.\frac{\partial E_m}{\partial V}\right|_{N,T} e^{-\beta E_m}\right) = -\left\langle \frac{\partial E}{\partial V}\right\rangle_{N,T} \tag{5.50}$$

$$S = -\left.\frac{\partial A}{\partial T}\right|_{N,V} = kT\left.\frac{\partial \ln Z}{\partial T}\right|_{N,V} + k\ln Z = k(\beta\langle E\rangle + \ln Z) \tag{5.51}$$

We already know the expression of the internal energy $U = -\left.\frac{\partial \ln Z}{\partial \beta}\right|_{N,V}$, from which the heat capacity at constant volume is obtained:

$$C_v = \left.\frac{\partial U}{\partial T}\right|_{N,V} = \left.\frac{\partial U}{\partial \beta}\frac{\partial \beta}{\partial T}\right|_{N,V} = \frac{1}{kT^2}\left.\frac{\partial^2 \ln Z}{\partial \beta^2}\right|_{N,V} = k\beta^2\left.\frac{\partial^2 \ln Z}{\partial \beta^2}\right|_{N,V} \tag{5.52}$$

We can also express entropy in terms of probability only. Let's start with the operative definition of probability:

$$P_n = \frac{e^{-\beta E_n}}{Z} \iff \ln P_n = -\beta E_n - \ln Z \tag{5.53}$$

We can use P_n to determine the average value of energy:

$$\beta\langle E\rangle = \sum_n P_n \beta E_n = \sum_n P_n(-\ln P_n - \ln Z) = -\sum_n P_n \ln P_n - \ln Z \sum_n P_n = -\sum_n P_n \ln P_n - \ln Z \tag{5.54}$$

Therefore

$$S = k(\beta\langle E\rangle + \ln Z) = k\left(-\sum_n P_n \ln P_n - \cancel{\ln Z} + \cancel{\ln Z}\right) = -k\sum_n P_n \ln P_n \tag{5.55}$$

We see that entropy depends solely on the probability values: it does not possess any associated microscopic magnitude; it is a **thermal quantity**. Eq. 5.55 leads to several important conclusions:

- If the system is in the ground state (at $T = 0$ K), and this state is unique, then $P_n = 1$ for this state and zero for all other states. The value of entropy for this state is zero, which is essentially the **Third Law of Thermodynamics**.
- As the temperature increases, the number of accessible energy states increases, and the values of P_n above the ground level are no longer zero. This leads to an increase in entropy. **Higher entropy implies a higher degree of statistical disorder within the system.**

Finally, we can express the partition function in terms of states or energy levels:

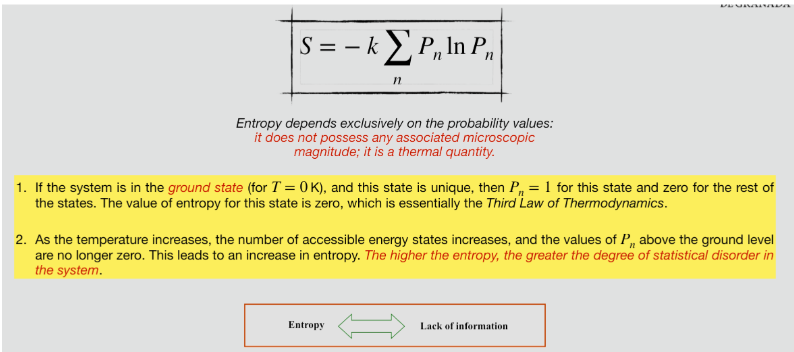
$$Z = \underbrace{\sum_m}_{\text{states}} e^{-\beta E_m} = \underbrace{\sum_n}_{\text{energy levels}} \sum_{[j]_n} e^{-\beta E_n} = \sum_n e^{-\beta E_n} \sum_{[j]_n} 1 = \sum_n \Omega(E_n) e^{-\beta E_n} \tag{5.56}$$

where $[j]_n$ indicates the number of states with energy level compatible with n. We also remember that

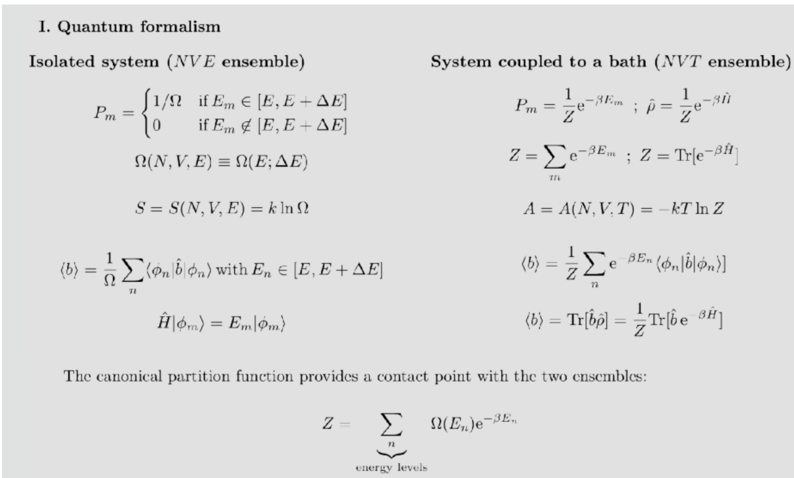
$$P_m = \frac{1}{Z} e^{-\beta E_m} \tag{5.57}$$

and

$$\langle B \rangle = \frac{1}{Z} \sum_m B_m e^{-\beta E_m} \tag{5.58}$$



1.2 Partition Function. Classical and Quantum Formalism



II. Classical formalism

Isolated system (NVE ensemble)

$$\rho = \frac{1}{N!h^s}\frac{1}{\omega(E)}\delta(E - H(\mathbf{q},\mathbf{p}))$$

$$\omega(E) = \frac{1}{N!h^s}\int \delta(E - H(\mathbf{q},\mathbf{p}))d\mathbf{q}\,d\mathbf{p}$$

$$\Omega(E) = \omega(E)\Delta E \;\; ; \;\; S = k \ln \Omega$$

$$\langle b \rangle = \frac{1}{N!h^s\omega(E)}\int b(\mathbf{q},\mathbf{p})\delta(E - H(\mathbf{q},\mathbf{p}))d\mathbf{q}\,d\mathbf{p}$$

System coupled to a bath (NVT ensemble)

$$\rho(\mathbf{q},\mathbf{p}) = \frac{1}{N!h^s}\frac{1}{Z}\mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})}$$

$$Z = \frac{1}{N!h^s}\int \mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})}d\mathbf{q}\,d\mathbf{p}$$

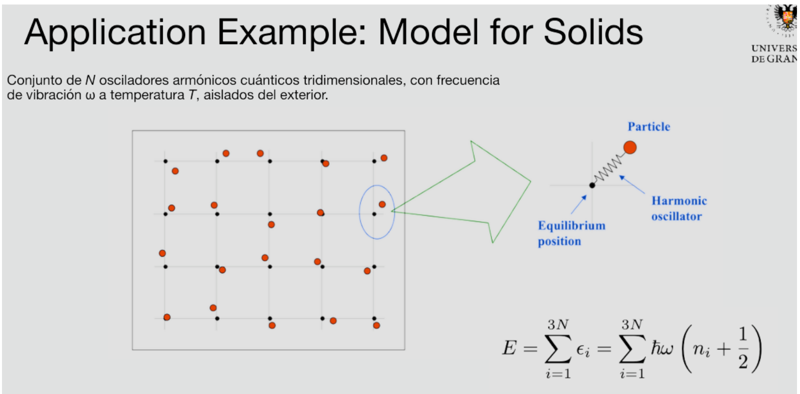
$$A = -kT \ln Z \;\; ; \;\; S = -k\int \rho(\mathbf{q},\mathbf{p}) \ln[N!h^s\rho(\mathbf{q},\mathbf{p})]d\mathbf{q}\,d\mathbf{p}$$

$$\langle b \rangle = \frac{1}{N!h^s}\frac{1}{Z}\int b(\mathbf{q},\mathbf{p})\mathrm{e}^{-\beta H(\mathbf{q},\mathbf{p})}d\mathbf{q}\,d\mathbf{p}$$

Again, the canonical partition function provides a contact point with the two ensembles:

$$Z = \int_0^\infty \mathrm{e}^{-\beta E}\omega(E)\,dE$$

SOME PROVES IN RAW



Let's consider a set of N quantum three-dimensional harmonic oscillators, with vibration frequency ω at temperature T isolated from the surroundings, as shown in Fig. 5.3. Our aim is calculating the canonical partition function Z and, once the expression for Z is obtained, the thermodynamic properties. To this end, we make use of the Einstein solid, which consists of $3N$ distinguishable and independent oscillators of equal frequency, ω. In the Einstein crystal model, the partition function can then be derived by treating the system as a collection of $3N$ independent quantum harmonic oscillators, corresponding to the vibrational degrees of freedom of the N atoms in the crystal.

To obtain Z, we should first write down the total energy of this system, which is given by

$$E = \sum_{i=1}^{3N} \epsilon_i = \sum_{i=1}^{3N} \hbar\omega \left(n_i + \frac{1}{2}\right) \tag{5.62}$$

with $n_i = 0, 1, \ldots \infty$. It follows that the partition function reads

$$Z = \sum_{\{E\}} e^{-\beta E} = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} \cdots \sum_{n_{3N}=0}^{\infty} e^{-\beta\hbar\omega \sum_{i=1}^{3N}(n_i+\frac{1}{2})} \tag{5.63}$$

which gives

$$Z = \underbrace{\sum_{n_1=0}^{\infty} e^{-\beta\hbar\omega(n_1+\frac{1}{2})}}_{Z_1} \underbrace{\sum_{n_2=0}^{\infty} e^{-\beta\hbar\omega(n_2+\frac{1}{2})}}_{Z_2} \cdots \underbrace{\sum_{n_{3N}=0}^{\infty} e^{-\beta\hbar\omega(n_{3N}+\frac{1}{2})}}_{Z_{3N}} = (Z_1)^{3N} \tag{5.64}$$

$$Z_1 = \sum_{n=0}^{\infty} e^{-\beta\hbar\omega(n+\frac{1}{2})} = e^{-\beta\hbar\omega/2} \sum_{n=0}^{\infty} e^{-\beta\hbar\omega n} = e^{-\beta\hbar\omega/2} \frac{1}{1-e^{-\beta\hbar\omega}} \tag{5.65}$$

where we have made use of the property $\sum_{n=0}^{\infty} a^n = 1/(1-a)$ for $a < 1$ and $a = e^{-\beta\hbar\omega}$. It follows that

$$Z_1 = \frac{1}{2\sinh(\beta\hbar\omega/2)} = \frac{1}{2\sinh(\hbar\omega/2kT)} \tag{5.66}$$

and

$$Z = \left(\frac{1}{2\sinh(\hbar\omega/2kT)}\right)^{3N} \tag{5.67}$$

The Helmholtz free energy reads

$$A = -kT\ln Z = 3NkT\ln\left[2\sinh\left(\frac{\hbar\omega}{2kT}\right)\right] \tag{5.68}$$

which is, as it is easy to check, an extensive variable. Let's now calculate entropy and internal energy U:

$$S = -\left.\frac{\partial A}{\partial T}\right|_{N,V} = -3Nk\left.\frac{\partial\{T\ln[2\sinh(\hbar\omega/2kT)]\}}{\partial T}\right|_{N,V} = -3Nk \Big(\ln[2\sinh(\beta\hbar\omega/2)] - \frac{\beta\hbar\omega}{2}\coth(\beta\hbar\omega \tag{5.69}$$

$$U = -\left.\frac{\partial\ln Z}{\partial\beta}\right|_{N,V} = \left.\frac{\partial\{3N\ln[2\sinh(\beta\hbar\omega/2)]\}}{\partial\beta}\right|_{N,V} = 3N\frac{\cosh(\beta\hbar\omega/2)}{\sinh(\beta\hbar\omega/2)}\frac{\hbar\omega}{2} = \frac{3N\hbar\omega}{2}\coth\left(\frac{\beta\hbar\omega}{2}\right) \tag{5.70}$$

We also see that for very large temperatures, we recover the result that we obtained in the microcanonical ensemble, that is

$$\lim_{T \to \infty} U = 3NkT \tag{5.71}$$

Other thermodynamic functions can be calculated by applying the same strategy.

1.3 Application to Ideal Classical Systems: Boltzmann Statistics

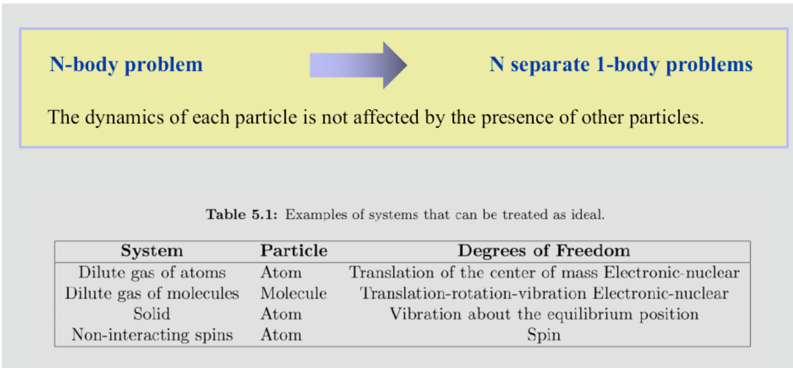


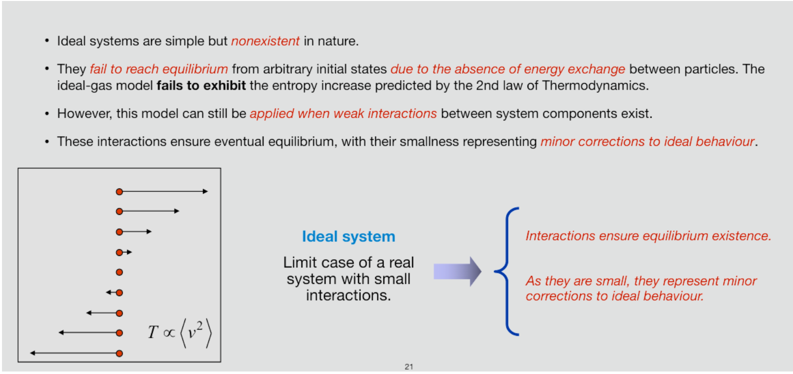
Table 5.1: Examples of systems that can be treated as ideal.

System	Particle	Degrees of Freedom
Dilute gas of atoms	Atom	Translation of the center of mass Electronic-nuclear
Dilute gas of molecules	Molecule	Translation-rotation-vibration Electronic-nuclear
Solid	Atom	Vibration about the equilibrium position
Non-interacting spins	Atom	Spin

What do we mean by ideal system? It's the application of Statistical Physics to a system formed by N particles, with $N \gg 1$. The resolution of the Schrödinger equation of the system, although necessary to know $E_j(N, V)$, is, in general, a prohibitive task. However, there exist a significant number of systems of great physical interest, whose Hamiltonian can be separated into independent terms:

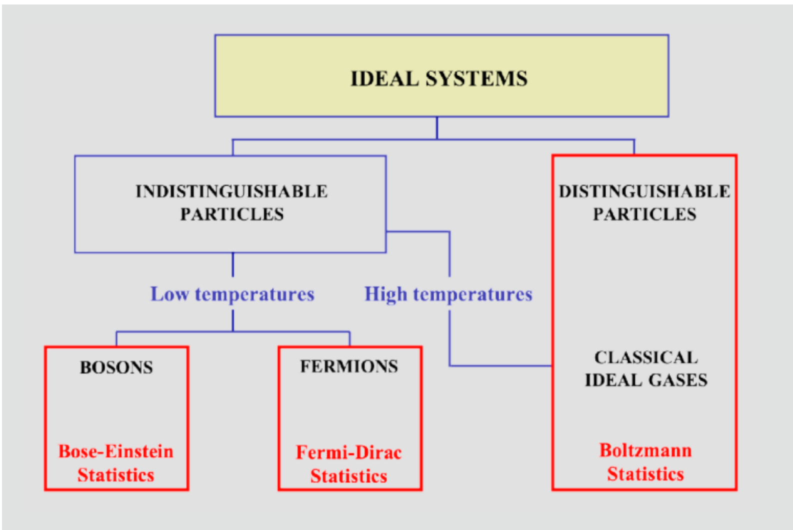
$$H(N) = \sum_{j=1}^{N} H_j \longrightarrow E(N) = \sum_{j=1}^{N} \epsilon_j \tag{5.72}$$

The **ideal system** is a set where H_j involves a reduced set of degrees of freedom that do not interact with the rest, consisting of **mutually independent** particles. Consequently, the problem of N bodies can be separated into N individual one-body problems. The dynamics of each particle is not affected by the presence of other particles - some examples of this kind are included in Table 5.1.



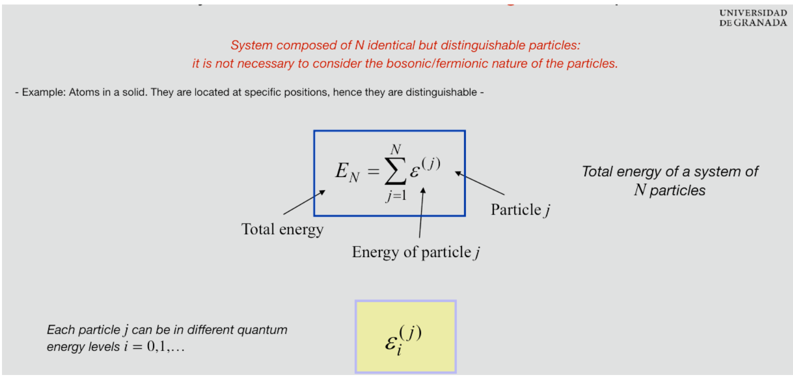
The Second Law states that for an isolated system, entropy tends to increase over time, leading to equilibrium. However, in an ideal system, (1) if particles do not interact, their individual energies remain constant; (2) there is no mechanism for redistributing energy among the components of the system; and (3) without redistribution, a system that starts out in a non-equilibrium state (*e.g.*, with different regions having different energy distributions) will remain in that state indefinitely. Since the Second Law relies on microscopic interactions driving the system toward a more probable (higher entropy) state, a system without interactions fails to obey this principle. However, in real systems, even weak interactions restore thermalisation and entropy growth over time.

An ideal system, as typically defined in physics, consists of non-interacting particles. If there is no interaction between particles, then no energy or momentum exchange can occur between them. This can lead to situations where the system does not evolve toward equilibrium from an arbitrary initial state, which



appears to contradict the Second Law of Thermodynamics[5]. Despite these limitations, this model remains very useful when weak interactions between system components exist. These interactions allow eventual equilibration, with their smallness representing minor corrections to ideal behaviour. With reference to Fig. 5.4, we will now explore the characteristics of ideal systems of distinguishable and indistinguishable particles.

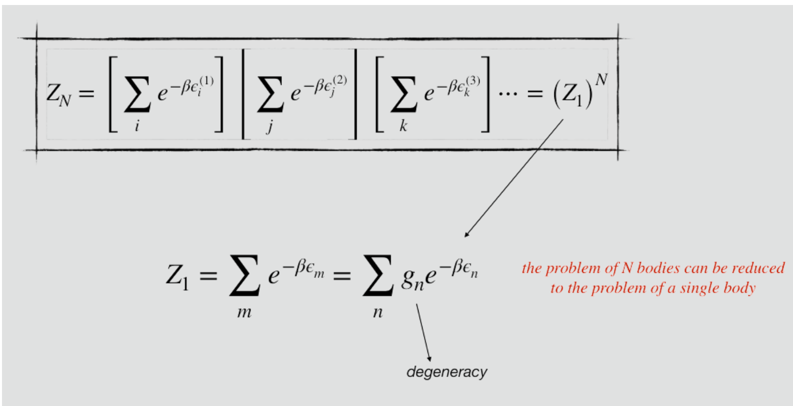
1.3.1 Case 1. Ideal systems of identical and distinguishable particles



A system composed of N identical but distinguishable particles does not require consideration of the bosonic/fermionic nature of the particles. For example, atoms in a solid fit this description. They are localised at specific positions, thus, they are distinguishable. The total energy of this system can be written as:

$$E = \sum_{\alpha=1}^{N} \epsilon_{\xi}^{(\alpha)} = \epsilon_i^{(1)} + \epsilon_j^{(2)} + \epsilon_k^{(3)} + \ldots \tag{5.73}$$

where $\alpha = 1, 2, \ldots, N$ labels a specific particle, whose quantum state is indicated by $\xi = i, j, k, \ldots$.



In particular, Z can be calculated as the sum over all configurations compatible with the system total energy:

$$Z_N = \sum_{\{E\}} e^{-\beta E} \tag{5.74}$$

Summing for all E values is equivalent to summing over any configuration of the quantum numbers i, j, k, etc.

$$Z_N = \sum_{i,j,k,\ldots} e^{-\beta[\epsilon_i^{(1)} + \epsilon_j^{(2)} + \epsilon_k^{(3)} + \ldots]} \tag{5.75}$$

We can solve these sums by breaking down the argument of the sum into products of exponentials and grouping:

$$Z_N = \left[\sum_i e^{-\beta\epsilon_i^{(1)}}\right]\left[\sum_j e^{-\beta\epsilon_j^{(2)}}\right]\left[\sum_k e^{-\beta\epsilon_k^{(3)}}\right]\ldots \tag{5.76}$$

and in total, we have N factors. Since the particles are identical, the energy spectrum $\{\epsilon_i\}$ is the same for all; therefore, these N factors are identical. Let's define the partition function of a single particle:

$$Z_1 = \sum_m e^{-\beta\epsilon_m} \tag{5.77}$$

where m represents the number of states. Because particles are ideal and distinguishable, then

$$Z_N = (Z_1)^N \tag{5.78}$$

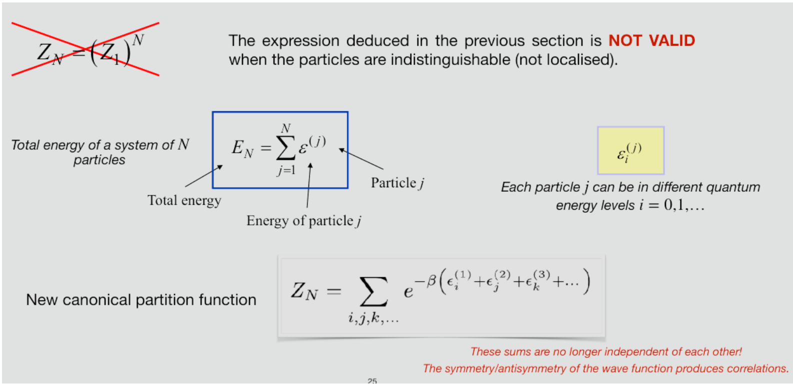
if there is degeneracy[6], then Z_1 is better expressed as a sum over energy levels the form

$$Z_1 = \sum_n g_n e^{-\beta\epsilon_n} \tag{5.79}$$

where g_n is the degeneracy of the energy level ϵ_n of a particle. In conclusion, for an ideal system with N distinguishable particles, **the problem of N bodies can be reduced to the problem of a single body**. If there are more species, for instance there are N_A particles of type A and N_B particles of type B, then the resulting partition function reads

$$Z_N(N_A, N_B, T, V) = (Z_1^{(A)})^{N_A}(Z_1^{(B)})^{N_B} \tag{5.80}$$

1.3.2 Case 2. Ideal systems of identical and indistinguishable particles



The expression derived in the previous section, that is $Z_N = (Z_1)^N$, is no longer valid when particles are indistinguishable (non-localised). The total energy of the ideal system results from the sum of distinct, autonomous terms.

$$E_N = \sum_{j=1}^{N} \epsilon^{(j)} \tag{5.81}$$

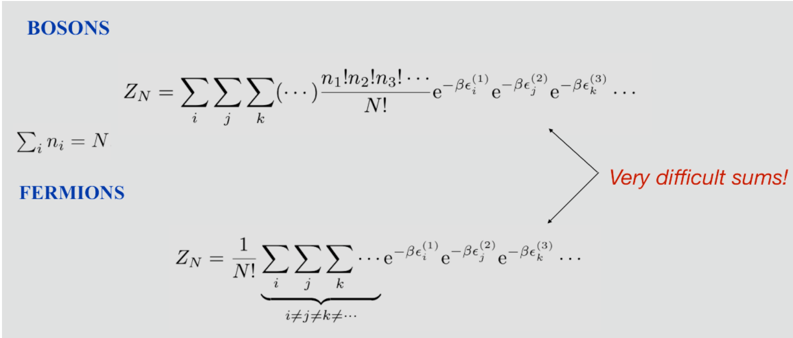
Ideal conditions allow for the definition of independent quantum states for each particle:

$$E_N = \epsilon_i^{(1)} + \epsilon_j^{(2)} + \epsilon_k^{(3)} + \ldots \tag{5.82}$$

and the new canonical partition function reads

$$Z_N = \sum_{i,j,k,\ldots} e^{-\beta(\epsilon_i^{(1)}+\epsilon_j^{(2)}+\epsilon_k^{(3)}+\ldots)} \tag{5.83}$$

These sums are no longer independent of each other. The symmetry/antisymmetry of the wave function generates correlations. There is a need to distinguish between bosons and fermions.



1. Bosons.

- Example 1. There are 5 bosons in the states $\{\epsilon_a, \epsilon_b, \epsilon_c, \epsilon_d, \epsilon_e\}$. Therefore there are $W = 5!$ ways to arrange the 5 particles.
- Example 2. There are 5 bosons in the states $\{\epsilon_a, \epsilon_a, \epsilon_a, \epsilon_b, \epsilon_b\}$. Therefore there are $W = \frac{5!}{3!2!} = 10$ ways to arrange the 5 particles.
- Example 3. There are 5 bosons in the states $\{\epsilon_a, \epsilon_a, \epsilon_a, \epsilon_a, \epsilon_b\}$. Therefore there are $W = \frac{5!}{4!1!} = 5$ ways to arrange the 5 particles.

In general, if there are n_1 bosons in state ϵ_1, n_2 in state ϵ_2, n_3 in state $\epsilon_3, \dots,$ then $W = \frac{N!}{n_1!n_2!n_3!\dots}$ with $N = n_1 + n_2 + n_3 + \dots$. Consequently, for bosons, we have:

$$Z_N = \sum_i \sum_j \sum_k (\dots) \frac{n_1!n_2!n_3!\dots}{N!} e^{-\beta\epsilon_i^{(1)}} e^{-\beta\epsilon_j^{(2)}} e^{-\beta\epsilon_k^{(3)}} \dots \tag{5.84}$$

Due to the restriction $\sum_i n_i = N$, the above summation is indeed very complicated.

2. Fermions.
As far as fermions are concerned, one can apply **Pauli's Exclusion Principle**. It follows that either are the states empty or they are occupied by a single particle:

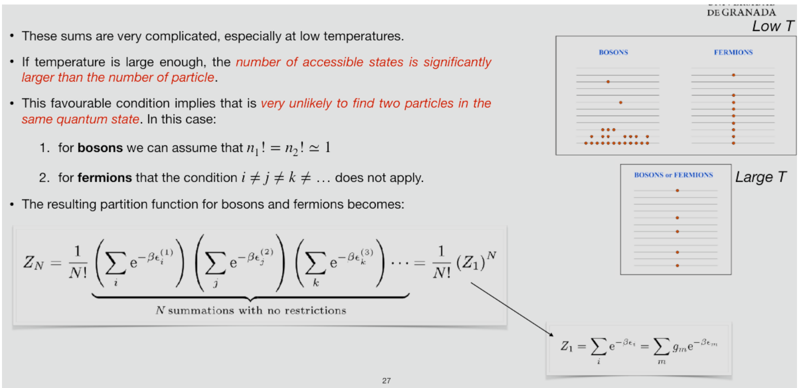
$$n_i = \{0, 1\} \to n_i! = 1 \tag{5.85}$$

and the partition function reads

$$Z_N = \frac{1}{N!} \sum_i \sum_j \sum_k \dots e^{-\beta\epsilon_i^{(1)}} e^{-\beta\epsilon_j^{(2)}} e^{-\beta\epsilon_k^{(3)}} \dots \tag{5.86}$$

Figure 5.5: At low temperatures, the thermal energy (kT) is small, making it very difficult for a particle to get access to high energy levels. Instead, most particles

remain in low-energy quantum states. Therefore, the number of accessible states is of the same order or smaller than the number of particles.

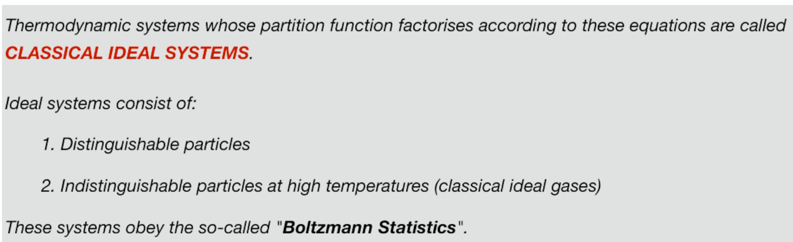


In general, the above summations for bosons and fermions are very complicated, especially so at low temperatures (see Fig. 5.5). Nevertheless, if temperature is large enough, the number of accessible states is significantly larger than the number of particles (see Fig. 5.6). This favourable condition implies that is very unlikely to find two particles in the same quantum state. Therefore, for bosons we can assume that $n_1! = n_2! \simeq 1$ and for fermions that the condition $i \neq j \neq k \neq \ldots$ does not apply. The resulting partition function for both bosons and fermions is then modified as follows

$$Z_N = \frac{1}{N!} \underbrace{\left(\sum_i e^{-\beta\epsilon_i^{(1)}}\right)\left(\sum_j e^{-\beta\epsilon_j^{(2)}}\right)\left(\sum_k e^{-\beta\epsilon_k^{(3)}}\right)\ldots}_{N \text{ summations with no restrictions}} = \frac{1}{N!}(Z_1)^N \tag{5.87}$$

where

$$Z_1 = \sum_i e^{-\beta\epsilon_i} = \sum_m g_m e^{-\beta\epsilon_m} \tag{5.88}$$



Thermodynamic systems whose partition function factorises according to these equations are known as **classical ideal systems**. Ideal systems consist of:

1. Distinguishable particles.

2. Indistinguishable particles at high temperatures (classical ideal gases).

These systems follow the so-called *Boltzmann Statistics*. A summary of their thermodynamic properties is provided in Fig. 5.7.

$$Z_N = \frac{1}{N!}(Z_1)^N \qquad A(T,V,N) = -kT\ln Z_N = -NkT\left[\ln\left(\frac{Z_1}{N}\right)+1\right]$$

$$U = NkT^2\left(\frac{\partial \ln Z_1}{\partial T}\right)_V$$

$$S = NkT\left(\frac{\partial \ln Z_1}{\partial T}\right)_V + kN\left[\ln\left(\frac{Z_1}{N}\right)+1\right]$$

$$P = NkT\left(\frac{\partial \ln Z_1}{\partial V}\right)_{T,N}$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = NkT^2\left(\frac{\partial^2 \ln Z_1}{\partial T^2}\right)_V + 2NkT\left(\frac{\partial \ln Z_1}{\partial T}\right)_V$$